

Electrical Power System (EPS) Design for 1U CubeSat Model

This report details the design of the Electrical Power System (EPS) for your 1U (10x10cm) CubeSat model, focusing on the integrated Power Distribution and Solar Charge Controller PCB. The design incorporates an STM32 microcontroller for control, a reaction wheel motor, and a Bluetooth Low Energy (BLE) communication link for telemetry to a laptop ground station.

1. System Requirements and Preliminary Power Budget

The design is based on a single Li-ion battery (3.7V nominal) and requires two regulated power rails: **3.3V** for logic and **5.0V** for the motor.

1.1. Estimated Power Consumption

A preliminary power budget was established to guide component selection and battery sizing. The nominal power consumption of the model is estimated to be well within the typical range for a 1U CubeSat [1].

Component	Operating Voltage (V)	Estimated Current (mA)	Estimated Power (mW)	Notes
STM32 (Active)	3.3	20	66	General-purpose active mode.
BLE Module (Active)	3.3	15	50	During transmission.
Reaction Wheel (Nominal)	5.0	100	500	Based on commercial low-power reaction wheels [2].
Sensors/Other (Sensing ICs)	3.3	10	33	Conservative estimate for monitoring hardware.
Power Conversion Loss (15%)	N/A	N/A	100	Estimated loss from DC/DC converters.
TOTAL ESTIMATED NOMINAL POWER	N/A	N/A	~750 mW	Total power draw is approximately 0.75 W .

1.2. Battery Sizing

A single **18650 Li-ion cell** is recommended. To ensure a minimum of **2 hours of operation** without solar charging (simulating ground testing or eclipse), the required capacity is approximately 405 mAh. A standard 18650 cell with a capacity of **2,000 mAh to 3,500 mAh** provides a robust margin for the model.

2. Solar Charge Controller (MPPT)

To maximize the energy harvested from the silica solar panels, a dedicated Maximum Power Point Tracking (MPPT) solution is essential.

Recommended MPPT IC: STMicroelectronics SPV1040

The SPV1040 is a high-efficiency step-up converter with embedded MPPT, specifically designed for low-power solar applications.

Feature	Specification	Rationale
Type	High-efficiency Step-up Converter with MPPT	Maximizes power from small solar panels, even in low-light conditions.
Input Voltage	0.3 V to 5.5 V	Excellent for boosting very low voltages from small solar cells.
Output Voltage	Configurable to 4.2 V	Matches the required maximum charge voltage for a single Li-ion cell.
Efficiency	Up to 95%	Ensures minimal energy loss during the charging process.

Note on Charging: The SPV1040 performs the MPPT function, but a separate Li-ion charger IC (e.g., STMicroelectronics L6924D or similar) is typically used in conjunction to handle the final Constant Current/Constant Voltage (CC/CV) charge profile and battery protection.

3. Power Distribution and Voltage Regulators

The requirement for two power rails (3.3V and 5.0V) from a single Li-ion battery (3.7V nominal) necessitates the use of high-efficiency switching regulators.

3.1. 3.3V Logic Rail (Buck-Boost)

The 3.3V rail powers the sensitive logic components (STM32, BLE, sensors). Since the battery voltage (3.0V to 4.2V) can be both above and below the required 3.3V, a **Buck-Boost Converter** is the most efficient solution.

Recommended Voltage Regulator: Texas Instruments TPS63020

The **TPS63020** is a highly efficient, single-inductor buck-boost converter, perfect for Li-ion battery applications where the input voltage crosses the output voltage. It can deliver up to 800 mA in boost mode and maintains up to 96% efficiency, which is crucial for maximizing battery life.

3.2. 5.0V Motor Rail (Boost)

The 5.0V rail powers the reaction wheel motor, which requires a higher voltage than the battery's nominal 3.7V. A dedicated **Boost Converter** is required to step up the voltage.

Recommended Voltage Regulator: Texas Instruments TPS61023

The **TPS61023** is a high-efficiency boost converter capable of supporting the required 500mA load at 5.0V. Its wide input voltage range (down to 0.5V) ensures that the motor rail remains stable even as the battery discharges.

4. Current and Voltage Sensing

To meet the requirement of measuring current and voltage and sending the telemetry to the ground station, an I²C-compatible current shunt monitor is recommended.

Recommended Sensing IC: Texas Instruments INA219

The **INA219** is a high-side, bidirectional current and power monitor that communicates via the I²C interface. It provides the STM32 with digital readings of:

- **Shunt Voltage** (used to calculate current)
- **Bus Voltage** (battery voltage)
- **Calculated Power**

Placing the INA219 between the battery and the power distribution stage allows the STM32 to monitor the total power consumption of the entire model, which is essential for power budgeting and health monitoring.

5. Conceptual EPS PCB Block Diagram

The final PCB design integrates these components as follows:

1. **Energy Harvesting:** Solar Panels $\rightarrow$ **SPV1040** (MPPT) $\rightarrow$ **L6924D/Similar** (Li-ion Charger)
2. **Energy Storage:** Charger $\rightarrow$ **18650 Li-ion Battery**
3. **Monitoring:** Battery $\rightarrow$ **INA219** (Current/Voltage/Power Monitor)
4. **Power Distribution:**
 - INA219 Output $\rightarrow$ **TPS63020** (3.3V Buck-Boost) $\rightarrow$ **STM32, BLE, Sensors**
 - INA219 Output $\rightarrow$ **TPS61023** (5.0V Boost) $\rightarrow$ **Reaction Wheel Motor**

6. Summary of Recommended Components

Subsystem	Component	Type	Key Specification
Battery	18650 Li-ion Cell	Storage	3.7V Nominal, 2000+ mAh Capacity
Solar MPPT	STMicroelectronics SPV1040	MPPT Controller	V _{in} : 0.3V-5.5V, V _{out} : up to 5.2V
3.3V Regulator	Texas Instruments TPS63020	Buck-Boost Converter	V _{in} : 1.8V-5.5V, I _{out} : 800mA (Boost), up to 96% efficiency
5.0V Regulator	Texas Instruments TPS61023	Boost Converter	V _{in} : 0.5V-5.5V, I _{out} : Supports 500mA @ 5V, up to 96% efficiency
Sensing	Texas Instruments INA219	I ² C Current/Power Monitor	High-side sensing, digital output, measures V, I, P
Control	STM32 Microcontroller	Control/Telemetry	Handles I ² C communication with INA219 and BLE transmission.

References

[1] Tadanki, A. *Closing the Power Budget Architecture for a 1U CubeSat*. Georgia Institute of Technology. [2] WITTENSTEIN cyber motor GmbH. *Reaction wheel - Technical data*. [Online]. Available: <https://cyber-motor.wittenstein.de/en-en/reaction-wheel-satellite/>